

Pharmacoeconomic Evaluation of a Pharmacist-Managed Hypertension Clinic

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Study Objective. To measure clinical, economic, and humanistic outcomes associated with a pharmacist-managed hypertension clinic compared with physician-managed clinics.

Design. Prospective, randomized, comparative study.

Setting. Managed care organization.

Patients. A total of 330 patients with mild-to-moderate essential hypertension.

Intervention. Hypertension care provided by either the pharmacist-managed hypertension clinic or physician-managed general medical clinics.

Measurements and Main Results. Baseline and 6-month evaluations consisted of systolic and diastolic blood pressure measurements, a short-form health survey, and collection of health care utilization information. After treatment, blood pressure measurements were significantly lower ($p < 0.001$) in the pharmacist-managed hypertension clinic group than in the physician-managed clinic group. Patient satisfaction was significantly higher in the hypertension clinic group. Total costs for the hypertension clinic group were not different from those of the physician-managed clinic group (\$242.46 vs \$233.20, $p = 0.71$), but cost:effectiveness ratios were lower in the hypertension clinic group (\$27 vs \$193/mm Hg for systolic blood pressure readings, and \$48 vs \$151/mm Hg for diastolic blood pressure readings).

Conclusion. In a hypertension clinic, pharmacists can be a cost-effective alternative to physicians in management of patients, and they can improve clinical outcomes and patient satisfaction.

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Clinical pharmacists are an integral part of the health care system and seek to improve quality of pharmaceutical care by providing supportive services to health professionals and patients alike.^{1–4} Pharmacists have made significant contributions to improved patient care in

primary care clinics^{3–13} and in other clinical settings as well.^{14–23}

Pharmacists have extensive knowledge about drugs regarding their therapeutic administration, cost, and adverse effects and can help physicians choose appropriate drugs while maximizing clinical benefits and reducing the number of negative outcomes. Hypertension is one area in which pharmacists have demonstrated a positive impact in helping physicians regulate drug therapy. Several studies have documented the beneficial effect a pharmacist can have in a clinic: reducing drug costs while improving blood pressure control; minimizing potential drug

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interactions, hospitalizations, and emergency room visits; and improving patient perception of quality of health care provided.²³⁻³²

According to the National Education Program Working Group report on management of patients with hypertension and high blood cholesterol,³³ hypertension can be satisfactorily controlled in virtually all patients. The report states that the focal point should not be whether high blood pressure can be reduced, but whether it can be reduced at a reasonable cost with minimal adverse effects. Some studies have evaluated costs associated with treatment of hypertension³⁴⁻³⁷; others have evaluated quality of life in patients with hypertension.³⁸⁻⁴⁰ However, little data exist on the cost-effectiveness of a clinical pharmacist assisting in the management of patients with hypertension and how this affects patients' perceived satisfaction with their health care.

Methods

Study Design

This was a randomized, prospective, comparative study conducted at a hypertension clinic and general medicine clinics within a managed care facility. The study was approved by the institutional review board; all enrolled patients provided written informed consent.

Patients

Physicians practicing in internal medicine or family practice clinics were contacted to determine their interest in the study. Interested physicians gave consent for their patients to be enrolled. Patients who met eligibility criteria were identified from the sites' pharmacy databases, and letters explaining the study were mailed to these patients.

Interested patients were prescreened over the telephone to verify that they met inclusion criteria. Patients had to be at least 18 years old, be diagnosed with essential hypertension, be a member of the managed care organization for at least 1 year, and fill prescriptions at the managed care organization's pharmacies. The patients had to be taking the targeted antihypertensive drugs nifedipine, verapamil, captopril, diltiazem, clonidine, terazosin, propranolol, or lisinopril, or be taking at least three prescription antihypertensive drugs.

Patients were excluded from participation if they had secondary hypertension (secondary to

drugs or comorbid diseases including but not limited to chronic renal disease, malignancies, Cushing's syndrome, primary aldosteronism, or aortic coarctation), if they had significant end-organ disease (such that hospitalization was likely within the next few months), or if they had a baseline blood pressure higher than 200 mm Hg (systolic) or 105 mm Hg (diastolic).

If eligible, patients were randomly assigned to one of two groups. In the pharmacist-managed hypertension clinic, a clinical pharmacist managed the treatment of patients, who made up the experimental group. Physicians were contacted and provided consent for any therapeutic changes but were asked not to adjust drug therapy unless a lack of intervention would be dangerous for the patient. In the physician-managed clinic, physicians managed the treatment of patients independently with no pharmacy intervention; this was the control group.

Data Collection

Two evaluations were required for all patients: one at baseline and one at the final visit 6 months later. The pharmacist or physician could schedule additional clinic appointments at his discretion. Data from these visits were not included in the evaluation of blood pressure control but were included in the analyses for health care utilization.

Patients randomly assigned to the pharmacist-managed hypertension clinic group were counseled by the clinical pharmacist. The pharmacist informed the patients that an effort would be made to decrease the number of drugs they took for hypertension or to alter their therapy by administering more appropriate or less expensive drugs to achieve similar or improved blood pressure control. The pharmacist determined the most appropriate antihypertensive regimen for the patient and ordered laboratory tests as needed. The pharmacist also provided education on nonpharmacologic ways to control blood pressure.

Patients randomized to the physician-managed clinic group were referred back to their primary care provider for hypertension treatment. These patients received no intervention, and physicians treated them in the customary manner.

In both groups at the baseline evaluation, a patient's sitting blood pressure was measured with a Datascope Accutorr automated sphygmomanometer (Datascope Corporation,

Table 1. Data Obtained from Patients at Baseline Visit

Variable	Hypertension Clinic (n=164)	Physician Clinic (n=166)	p Value
Age, yrs (mean $\pm$ SD)	61.95 $\pm$ 11.4	61.71 $\pm$ 11.3	0.85
Weight, lbs (mean $\pm$ SD)	192.12 $\pm$ 43.4	190.31 $\pm$ 52.2	0.74
M/F	92/72	77/90	0.07
No. of years with hypertension (mean $\pm$ SD)	15.85 $\pm$ 10.1	14.40 $\pm$ 10.9	0.21
Newly diagnosed hypertension (no. of pts)	7	5	0.57
No. of prescriptions/patient (mean $\pm$ SD)	2.18 $\pm$ 1.1	2.06 $\pm$ 1.0	0.33
Concurrent disease (no. of pts)	98	95	0.74
Smoker (no. of pts)	15	9	0.18
Alcohol consumer (no. of pts)	12	9	0.47
Mean blood pressure (mm Hg)			
Systolic	144.23 $\pm$ 18.4	142.91 $\pm$ 18.0	0.51
Diastolic	82.79 $\pm$ 11.2	82.13 $\pm$ 11.4	0.59
Not evaluable (no. of pts)			
Dropout	11	11	0.71
Protocol violation	3	3	0.95
Lost to follow-up	8	16	0.23

Montvale, NJ). Two readings were taken for each patient, and the average of the two readings was recorded. Medical records were reviewed to assess all blood pressure measurements obtained at each visit.

To assess health-related quality of life, patients completed the short-form health survey (SF-36)⁴¹ at baseline. The SF-36 asks questions about general physical, social, and mental health and reports results for eight domains: physical functioning, role-physical, bodily pain, general health, vitality, social functioning, role-emotional, and mental health. The answers were collected and scored based on the SF-36 manual and interpretation guide.⁴¹

At the final evaluation, patients again had their sitting blood pressure measured and completed the SF-36. In addition, data on resource utilization were collected. For resource data, the number of emergency room visits, hospitalizations, and scheduled clinic visits were collected. For emergency visit information to be included, the primary reason for the visit had to be related to blood pressure. Data were collected for clinic visits only when blood pressure was measured and the clinician provided an evaluation of the patient's hypertension. Clinic visits at which blood pressure was measured but no evaluation was provided in the progress notes were not included in the analysis of health care utilization.

Information about each patient's antihypertensive drugs was collected over the study period. All costs related to antihypertensive drugs were

included. Cost comparisons were conducted in separate analyses using two sets of drug costs: acquisition cost and cost based on average wholesale price.

Costs for emergency room visits, hospitalizations, and clinic visits were based on actual managed care costs for health care services at our facility based on 1998 costs. Clinic visit costs were calculated for each group based on salary of the provider (pharmacist or physician). Total costs consisted of aggregate managed care costs for any health care service provided plus drug costs for each patient. Clinical effectiveness was defined by reduction in blood pressure over the study period. Cost:effectiveness ratios were calculated using total costs/mm Hg of blood pressure decreased.

Statistical Analysis

A change in diastolic blood pressure was the primary dependent variable used to calculate sample size. Preliminary calculations given an α value of 0.05 and a standard deviation of 20 mm Hg would give a power of greater than 90% to detect a difference of 10 mm Hg between the two groups with 90 patients in each group. Calculations were determined using the BMDP Solo Power Analysis software (1992 version) (Los Angeles, CA).

Descriptive statistics were used for baseline demographic variables in both groups. Determination of normality was assessed with the Wilk's test. Continuous variables for between-group comparisons were analyzed with the

Table 2. Data Obtained from Patients at Final Visit

Variable	Hypertension Clinic	Physician Clinic	p Value
Blood pressure (mm Hg)			
Systolic	135.10 ± 15.3 ^a	141.66 ± 17.9	<0.001
Diastolic	77.65 ± 8.7	80.67 ± 10.2	0.004
Blood pressure decrease (mm Hg)			
Systolic	9.13 ± 17.1	1.32 ± 15.7	<0.001
Diastolic	5.14 ± 9.2	1.46 ± 10.1	<0.001
Clinic visits/patient (no.)	5.25 ± 2.3	1.41 ± 1.7	<0.001
Visited emergency room (no. of pts)	0	4.0	0.04
Prescriptions/patient (no.)	2.12 ± 1.1	2.20 ± 1.1	0.67

Student *t* test for independent samples, the Wilcoxon rank sum test if data were nonparametric, or the Welch's test in the event of heteroscedasticity. Within-group comparisons were analyzed with paired *t* tests or repeated measures analysis of variance as appropriate. Categorical data were compared using χ^2 analysis, the Fisher exact test, or appropriate contingency table analyses. Level of significance was set at an α of 0.05. All analyses were conducted using BMDP Dynamic Release software 7.0 (1993 version).

Results

Of the 1542 patients who responded to the telephone screening survey, 381 met the inclusion criteria and were randomly assigned to one of the treatment groups. Fifty-one patients were not included in the study for various reasons, such as physicians not wishing to have their patients participate, protocol violations, patient dropouts, and patients lost to follow-up. The remaining 330 patients (164 in the pharmacist-managed hypertension clinic, 166 in the physician-managed clinic) were included for analysis.

Baseline data are presented in Table 1. No statistically significant differences were noted between the groups. The pharmacist-managed hypertension clinic group had more men than the physician-managed clinic group, but the difference ($p=0.07$) was not statistically significant.

Blood Pressure

Within-group analyses for the pharmacist-managed hypertension clinic group revealed statistically significant decreases in both systolic and diastolic blood pressure, comparing baseline values with those at the final visit (Table 2).

From baseline to final visit, systolic blood pressure decreased from 144.23 mm Hg to 135.10 mm Hg ($p<0.001$) and diastolic blood pressure decreased from 82.79 mm Hg to 77.65 mm Hg ($p<0.001$). For the physician-managed clinic group, no statistically significant reductions were noted in the mean blood pressures during the study. The mean baseline systolic blood pressure was not different from the final visit systolic blood pressure (142.91 mm Hg vs 141.66 mm Hg, $p=0.31$). The final diastolic blood pressure was less than the baseline value, but the change was not statistically significant (82.13 mm Hg vs 80.67 mm Hg, $p=0.06$).

Between-group comparisons of blood pressure reductions over the 6 months showed that mean systolic blood pressure decreases between the pharmacist-managed hypertension clinic and physician-managed clinic groups were statistically different (9.13 mm Hg vs 1.32 mm Hg, $p<0.001$). Differences in mean diastolic blood pressure decreases also were statistically significant between the pharmacist-managed hypertension clinic and physician-managed clinic groups (5.1 mm Hg vs 1.46 mm Hg, $p<0.001$) (Figure 1).

Quality of Life

The results of changes in quality of life were evaluated and compared between and within groups (Table 3). High scores in any SF-36 domain indicate improved functioning for that particular domain. In our study, baseline between-group measurements indicated no statistically significant differences for any of the domains. At the end of the study, the only statistically significant difference was found in the role-physical domain, in which scores were significantly higher ($p=0.03$) in the pharmacist-managed hypertension clinic group than in the

Table 3. Quality-of-Life Data Obtained from Patients at Baseline and Final Visits

SF-36 Domain	Hypertension Clinic		Physician Clinic		p Value PC vs HC	
	Baseline	Final	Baseline	Final	Baseline	Final
Physical functioning	76.9 ± 24.1	78.2 ± 24.8	78.9 ± 23.7 ^a	75.8 ± 24.4	0.24	0.27
Role-physical	76.8 ± 36.3	78.5 ± 36.7	74.8 ± 36.0	70.7 ± 38.6	0.50	0.03
Bodily pain	71.0 ± 26.4	71.3 ± 26.0	69.1 ± 27.9	68.6 ± 25.9	0.61	0.31
General health	69.8 ± 19.2	70.4 ± 18.8	70.9 ± 17.8 ^a	66.6 ± 20.8	0.45	0.16
Vitality	61.7 ± 22.0	61.0 ± 22.3	61.8 ± 22.1	59.6 ± 22.8	0.92	0.55
Social functioning	86.1 ± 22.5	86.7 ± 20.8	83.2 ± 25.2	80.7 ± 26.5	0.30	0.06
Role-emotional	87.4 ± 30.1	86.0 ± 30.9	84.5 ± 33.1	79.7 ± 35.7	0.45	0.07
Mental health	80.0 ± 15.4	79.3 ± 15.0	79.1 ± 16.8	77.0 ± 17.0	0.72	0.51

HC = pharmacist-managed hypertension clinic; PC = physician clinic; SF-36 = short-form health survey.

^aStatistically significant ($p < 0.01$) changes between baseline and final values for the PC group.

physician-managed clinic group. Scores in social functioning and role-emotional also were higher in the pharmacist-managed hypertension clinic group than in the physician-managed clinic group but did not achieve statistical significance ($p = 0.06$ and $p = 0.07$, respectively).

Within-group comparisons of quality of life were conducted to determine whether individual patient scores within each group changed during the study. Results indicated that no significant changes occurred in quality of life in the pharmacist-managed hypertension clinic group when comparing baseline values with those at the final visit. In the physician-managed clinic group, significant ($p < 0.01$ for values at both baseline and final visit) reductions occurred in the physical functioning and general health domains. Overall, declines occurred in each domain for the physician-managed clinic group, but these did not reach statistical significance.

Health Care Utilization

Health care utilization was measured for both

groups over the 6-month study period. No patient was hospitalized during the study for reasons related to blood pressure. Four emergency room visits were reported in the physician-managed clinic group, none in the pharmacist-managed hypertension clinic group. One patient was admitted for evaluation of a headache. A subarachnoid hemorrhage was ruled out, and the attending physician noted that the patient's hypertension was poorly controlled. One patient was admitted for dizziness, and the physician noted changes in orthostatic blood pressure in the medical record. The other two patients were admitted with cardiac problems, and poor blood pressure control was noted in the medical record. The emergency room visits cost \$439/patient. The average number of clinic visits was significantly higher in the pharmacist-managed hypertension clinic group than the physician-managed clinic group (5.25 vs 1.41, $p < 0.001$) (Table 4).

Drug Utilization

The average number of antihypertensive drugs/patient at baseline taken by patients in the pharmacist-managed hypertension clinic and physician-managed clinic groups were not different (2.18 vs 2.06, $p = 0.33$). The number of drugs taken by the end of the study also were not different for the pharmacist-managed hypertension clinic and physician-managed clinic groups (2.12 vs 2.20, $p = 0.67$).

Cost-Effectiveness

No statistically significant differences occurred in mean drug cost/patient, cost of hospitalization, or total costs/patient between groups. Costs associated with clinic visits were higher in the

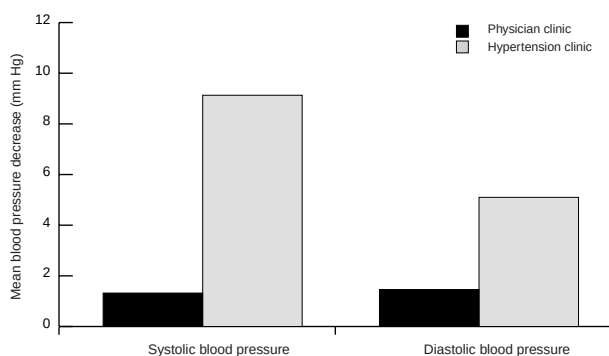


Figure 1. Mean decreases in blood pressure in the two patient groups.

Table 4. Health Care Utilization in Both Patient Groups

Variable	HC Group	PC Group	p Value
Drug cost/patient (\$)	111.79 ± 152.84	148.85 ± 229.51	0.08
Drug cost/patient based on AWP (\$)	390.24 ± 372.54	437.09 ± 402.61	0.28
Health care costs/patient (\$)			
Clinic visits	130.67 ± 58.75	73.51 ± 89.34	< 0.001
Emergency room visits	0	10.84	0.04
Hospitalizations	0	0	1.00
Total cost/patient (\$)	242.46 ± 169.16	233.20 ± 267.40	0.71
Total cost/patient based on AWP drug costs (\$)	520.91 ± 387.21	521.44 ± 438.30	0.99
Cost-effectiveness (\$/mm Hg)			
Marginal			
Systolic blood pressure	27	193	
Diastolic blood pressure	48	151	
Incremental			
Systolic blood pressure		1.18	
Diastolic blood pressure		2.51	

HC = pharmacist-managed hypertension clinic; PC = physician clinic; AWP = average wholesale price.

pharmacist-managed hypertension clinic than in the physician-managed clinic group (\$130.67 vs \$73.51/patient), which was statistically different ($p < 0.001$). Average cost/patient related to emergency room visits was lower in the pharmacist-managed hypertension clinic than in the physician-managed clinic group (\$0 vs \$10.84, $p < 0.04$).

In the pharmacist-managed hypertension clinic group, marginal cost:effectiveness ratios were lower for both systolic and diastolic blood pressures. The cost:effectiveness ratio for the pharmacist-managed hypertension clinic and physician-managed clinic groups was \$27 and \$93/mm Hg, respectively. The cost of decreasing the diastolic blood pressure 1 mm Hg was \$48 for the pharmacist-managed hypertension clinic group versus \$151 for the physician-managed clinic group. Incremental cost:effectiveness ratios for systolic and diastolic blood pressures were \$1.18 and \$2.51/mm Hg, respectively.

Discussion

The value of a clinical pharmacist assisting in the management of patients with hypertension has been documented.^{23-30, 32} However, few studies determined the cost-effectiveness of pharmacists in a randomized, prospective trial. Results of our study illustrate some interesting differences. At the end of the study, patients treated in the hypertension clinic had significantly lower blood pressures than those treated in the physician clinics. Interpretation of quality of life scores show only minor differences

between the groups; only the role-physical functioning domain indicated a statistical improvement in the pharmacist-managed hypertension clinic group. The real consequence of this finding remains to be seen because no differences may be perceptible in quality of life, the tool we used was not responsive enough to detect the changes, or differences may be detectable only after a longer period of follow-up than we used.

In general, more patients in the pharmacist-managed hypertension clinic group were satisfied than in the physician-managed clinic group. The reason for more satisfaction and improved blood pressure control in the pharmacist-managed hypertension clinic group probably is related to the amount of time the pharmacists versus the physicians spent with the patients. This is supported by the outcome data; the pharmacists scheduled almost 4 times as many clinic appointments as the physicians did. In addition, the pharmacists generally spent more time/patient/clinic visit. It would be interesting to know whether the clinical or quality of life outcomes would have been different had the number of clinic visits and time spent with patients been the same for the pharmacists and physicians.

We had expected that pharmacists practicing in the hypertension clinic would have been able to reduce drug costs because of their greater awareness in this area; however, the data show no differences in mean cost of antihypertensive drugs during the study. Therefore, one could speculate that by reducing drug costs or by

reducing the number of clinic visits and amount of time the pharmacist spends with each patient could contribute to further cost savings. Unfortunately, this also could result in less optimal blood pressure control, decreased patient satisfaction, or increased health care utilization costs. Therefore, a balance among these factors should be weighed carefully.

Although total costs were not different between groups, the clinical pharmacists demonstrated greater cost-effectiveness than the physicians. It is interesting to note that this improved cost-effectiveness was not due to cost reductions but rather to greater effectiveness.

We acknowledge a number of limitations with this study. It would have been more desirable to study only patients newly diagnosed with hypertension so that previous treatment would not affect outcome. Unfortunately, the number of newly diagnosed patients was too small to achieve the necessary sample size. Our sample included only those patients taking certain antihypertensive drugs, which are typically the more costly antihypertensive agents. Results may have been different had we included patients who were taking other commonly used drugs, such as diuretics or β -blockers. Conventional methods for determining which costs to analyze typically involve all costs incurred/patient, regardless of whether they can be attributed to the independent variables (because it is difficult to determine which costs are associated and which are not). We analyzed only those costs that could be attributed to blood pressure or its treatment. We felt we could accurately determine which costs were related to hypertension and its treatment and minimize the possibility of erroneously including costs not attributable to the variables under study. We used only two blood pressure readings to determine effectiveness: baseline and 6 months later. Numerous blood pressure readings may improve the quality of the data because one aberrant blood pressure reading is less likely to affect results. However, this would make interpretation more difficult because blood pressure may fluctuate during the follow-up period. Physicians with the physician-managed clinic group were internists or family practitioners, and the results should not be extrapolated to other physicians.

Results of the quality of life evaluation using the SF-36 indicated minor between-group differences. However, it should be noted that the SF-36 is a nonspecific instrument and may not be sufficiently responsive in assessing patients with

essential hypertension or who are asymptomatic. A sensitivity analysis was conducted to determine if the results differed based on whether drug acquisition costs or costs based on average wholesale price were used. Our results were not sensitive to these different cost structures. Another limitation is that the sponsoring organization would not allow for dissemination of itemized costs of goods and services. Thus, results may have been different had our study been conducted at an institution with significantly different cost structures for drugs and other health care resources. Our results indicate that pharmacists practicing in the hypertension clinic were more cost-effective than the physicians, but effectiveness in this study was determined only by reductions in blood pressure. Although it is reasonable to assume that reductions in blood pressure result in decreased morbidity and mortality, this study did not actually measure these end points because the follow-up period would require many years.

Conclusion

Results of this study indicate that in a hypertension clinic, pharmacists can be a cost-effective alternative to physicians in managing the treatment of patients. The benefits of having pharmacists assist in the management of these patients include improved clinical outcomes and enhanced patient satisfaction, as well as allowing physicians to spend more time treating other patients.

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